

Exercises for Chapter 4

Multiple Choice Questions

1. Which of the following statements about the normal distribution is *false*?

- (a) The normal distribution is symmetric about its mean.
- (b) Approximately 95% of the data in a normal distribution lie within one standard deviation of the mean.
- (c) The total area under the normal distribution curve is equal to 1.
- (d) The normal distribution is completely determined by its mean and standard deviation.



2. If a dataset is normally distributed with mean μ and standard deviation σ , the Z -score of an observation x is calculated as:

- (a) $Z = \frac{x - \mu}{\sigma}$
- (b) $Z = \frac{\mu - x}{\sigma}$
- (c) $Z = (x - \mu) \times \sigma$
- (d) $Z = (\mu - x) \times \sigma$

3. The standard normal distribution is a normal distribution with:

- (a) Mean of 0 and standard deviation of 1.
- (b) Mean of 1 and standard deviation of 1.
- (c) Mean of 0 and variance of 1.
- (d) Mean of 1 and variance of 0.

4. The empirical rule states that for a normal distribution, approximately what percentage of data falls within two standard deviations of the mean?

- (a) 68%
- (b) 95%
- (c) 99.7%
- (d) 100%

5. Which of the following is *not* a property of a binomial distribution?

- (a) Fixed number of trials.
- (b) Each trial has only two possible outcomes.
- (c) Trials are independent.
- (d) Probability of success varies from trial to trial.

6. In a geometric distribution, the random variable X represents:

- (a) The number of successes in a fixed number of trials.

- (b) The number of failures until the first success.
 - (c) The probability of success in each trial.
 - (d) The total number of trials.
7. **The mean and variance of a binomial distribution with parameters n and p are:**
- (a) Mean = np , Variance = $np(1 - p)$
 - (b) Mean = np , Variance = $n^2p(1 - p)$
 - (c) Mean = np , Variance = $np^2(1 - p)$
 - (d) Mean = $n(1 - p)$, Variance = $np(1 - p)$
8. **Which of the following statements is true about the normal approximation to the binomial distribution?**
- (a) It can be used when $np \geq 10$ and $n(1 - p) \geq 10$.
 - (b) It can be used for any binomial distribution regardless of n and p .
 - (c) It cannot be used when p is close to 0.5.
 - (d) It is exact and does not require a continuity correction.
9. **The negative binomial distribution models:**
- (a) The number of trials until the first success.
 - (b) The number of successes in a fixed number of trials.
 - (c) The number of trials until a fixed number of successes occur.
 - (d) The probability of k successes in n trials.
10. **In a right-skewed distribution, most Z -scores are:**
- (a) Positive
 - (b) Negative
 - (c) Zero
 - (d) Undefined

Exercises for Chapter 4: Distributions of Random Variables

1. The heights of adult men in a certain population are approximately normally distributed with a mean of 70 inches and a standard deviation of 3 inches. **What proportion of men are taller than 73 inches?**
2. Test scores on a college entrance exam are normally distributed with a mean of 500 and a standard deviation of 100. **What is the minimum score a student must achieve to be in the top 10% of test-takers?**
3. A student scores 85 on a test where the class mean is 75 with a standard deviation of 5. **What is the student's Z -score?**
4. Marathon finishing times are normally distributed with a mean of 240 minutes and a standard deviation of 30 minutes. **What percentile is a runner in if they finish in 195 minutes?**
5. For a normally distributed variable with mean 100 and standard deviation 15, **what value corresponds to the 75th percentile?**
6. IQ scores are normally distributed with a mean of 100 and a standard deviation of 15. **What IQ score corresponds to the top 2% of the population?**
7. The weights of apples in an orchard are normally distributed with a mean of 150 grams and a standard deviation of 10 grams. **Approximately what percentage of apples weigh between 140 grams and 160 grams?**
8. Exam scores are normally distributed with a mean of 78 and a standard deviation of 6. **What percentage of students scored between 72 and 90?**
9. The lengths of nails produced by a machine are normally distributed with a mean of 5 cm and a standard deviation of 0.1 cm. **What proportion of nails are shorter than 4.8 cm?**
10. The lifespan of a certain brand of light bulbs is normally distributed with a mean of 1,200 hours and a standard deviation of 100 hours. **What is the probability that a randomly selected bulb lasts between 1,050 and 1,350 hours?**
11. A fair die is rolled repeatedly until a six appears. **What is the probability that the first six appears on the third roll?**
12. A website has a 20% chance of crashing each time it is accessed. **What is the expected number of accesses before the site crashes?**
13. A student randomly guesses on each of 10 multiple-choice questions, each with 4 options. **What is the probability that they get exactly 3 questions correct?**
14. A batch of 1,000 items contains 2% defective items. **If a sample of 50 items is randomly selected, what is the probability that the sample contains at most one defective item?**
15. A fair coin is tossed 100 times. **What is the approximate probability that it lands heads more than 60 times?**
16. In a city, 30% of households own a dog. **If 200 households are randomly surveyed, what is the probability that at least 75 households own a dog?**